CRISTOBAL, HAILEY JADE
3/19/26

BSIT 2-1
Midterm Activity #3 - OOP
Basic ATM system

I.) COMPLETE JAVA PROGRAM

```
package one;

import java.util.Scanner;

public class ATMTransactionCounterCristobal {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        double balance = 8000.0;    // 8000 is the initial balance requirement
        int transactionCount = 0;
        int choice = 0;

        while (choice != 4) {    // while loop for continuous transactions
            System.out.println("\n===== ATM MENU ======");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit");
            System.out.println("3. Withdraw");
            System.out.println("4. Exit");
            System.out.print("Choice: ");
            choice = input.nextInt();

            switch (choice) {    // switch-case for menu selection
                case 1:
                    System.out.println("Balance = " + balance);
                    transactionCount++;
                    break;

                case 2:
                    System.out.print("Enter deposit: ");
                    double deposit = input.nextDouble();
                    balance += deposit;
                    System.out.println("New Balance = " + balance);
                    transactionCount++;
                    break;

                case 3:
                    System.out.print("Enter withdrawal: ");
                    double withdraw = input.nextDouble();

                    if (withdraw > 2000) {    // if-else for validating withdrawal
                        System.out.println("Exceeds maximum withdrawal");    // max : 2000
                    } else if (withdraw > balance) {
                        System.out.println("Amount exceeds balance");
                    } else {
                        balance -= withdraw;
                        System.out.println("New Balance = " + balance);
                    }
                    transactionCount++;
                    break;

                case 4:
                    System.out.println("Transaction count: " + transactionCount);    // Final count
                    System.out.println("Exit selected!");
                    break;
```

```
            default:
                System.out.println("Invalid choice!");
        }
    }
    input.close();
  }
}
```

II.) SAMPLE EXECUTION

Transaction 1:
Choice: 1
Output: Balance = 8000

Transaction 2:
Choice: 2
Deposit: 1000
Output: New Balance = 9000

Transaction 3:
Choice: 3
Withdraw: 2500
Output: Exceeds maximum withdrawal

Final Result:
Transaction count: 3

III.) SHORT EXPLANATION

The program implements a while loop to facilitate a persistent execution state, allowing the user to perform multiple transactions recursively until the termination condition (option 4) is met.

A switch-case structure is used as the primary control mechanism to map the user's integer input to specific functional blocks, such as balance inquiries, deposits, or withdrawals.

To ensure data integrity and enforce business logic, if-else statements are nested within the withdrawal case to validate that the requested amount neither exceeds the single transaction limit of 2,000 nor the current available balance.